eHealth NSW transforms public health system with the cloud

An eHealth NSW case study

New South Wales (NSW) Health's vision is to create a sustainable health system that delivers outcomes that matter to patients, is personalised, invests in wellness and is digitally enabled. In line with this vision, it is currently on a digital transformation journey to build an even stronger, more flexible, patient-centered health system for its citizens.

eHealth NSW plans, implements and supports information and communication technology (ICT) and digital capabilities across NSW Health by designing and implementing technology solutions to innovate and transform how healthcare is delivered to patients. With their passion for collaborating with clinicians to build industry-leading healthcare solutions, eHealth NSW is the digital centre of excellence for NSW Health.

From 2021 to 2022, eHealth NSW migrated 10 clinical applications, including mission critical workloads from its on-premises infrastructure onto the AWS Cloud. With AWS and its managed services, eHealth NSW has reduced manual operational activities, shortened the time to create new environments and achieved performance improvements for its applications. During the pandemic, eHealth NSW was also able to swiftly scale its video conferencing solution to support staff working remotely, clinicians and patient consultations.

Modernising eHealth NSW's infrastructure to save time and improve performance

As a ICT service provider for the state's public health system, eHealth NSW handles highly sensitive and personal data. Being on the AWS Cloud, eHealth NSW can take advantage of cloud-based security services that provide enhanced visibility, and auditing capabilities and controls. This is projected to save NSW Health hundreds of hours per year on manual and reactive operational activities. eHealth NSW uses AWS Security Hub and Amazon GuardDuty to automate security checks, implement continuous monitoring, and conduct early threat detection to minimise security risks.

Clinicians and researchers are also seeing improvements to NSW Health's various clinical applications since the move to the AWS Cloud. For example, the Enterprise Patient Records

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application has realised a 10-fold improvement in performance and a 70 percent reduction in critical incidents, resulting in faster access to clinical data for frontline clinicians. Additionally, eHealth NSW can implement enhancements to applications up to 50 percent faster, as they can now launch new environments in under four hours to conduct testing. In the past, this would have taken them six to eight weeks. NSW Health has already recognised US$16 million-worth of benefits due to avoided costs, reduction in capital and expenditure and improvements in productivity. Up to 144,000 collective hours of productivity have also been saved for frontline clinicians.

To support growing data needs, eHealth NSW also deployed Amazon FSx for NetApp ONTAP for its Enterprise Image Repository (EIR). The EIR, which is a centralised medical imaging store, currently holds over 1.6 petabytes of diagnostic images and reports, and its data is growing at a rate of 25 percent per annum. EIR also uses Amazon Relational Database Service (Amazon RDS) for high availability and encryption, and to automate time-consuming database administration tasks, such as software patching and backups.

"We worked closely with the AWS Professional Services team to modernise our infrastructure, allowing us to redirect our resources toward improving the clinical application features and delivering greater value to the state healthcare system," said Farhoud Salimi, Executive Director, Service Delivery at eHealth NSW. "Additionally, the AWS team has been very helpful in equipping our team with the resources and training we need to manage our new infrastructure."

Adapting to new communication needs during the pandemic

eHealth NSW's migration to the cloud also proved to be very timely during the COVID-19 pandemic, when the use of video conferencing increased by 18 times due to a surge in remote workers and telehealth. With the AWS Cloud, eHealth NSW was able to ensure ample network, server and storage capacity to maintain performance at scale.

NSW Health Pathology used Amazon Connect via the eHealth NSW cloud platform to build a new MVP COVID testing call centre within two weeks. At its peak, the call centre scaled to manage 40,000 tickets per week. To further ease public anxiety, NSW Health Pathology also set up an SMS bot using Amazon Pinpoint. This initially allowed patients with negative results to receive a notification within hours, instead of waiting up to 10 days, and was enhanced throughout the pandemic to include positive results and most recently results for Influenza A, B and RSV.

Facilitating upskilling of staff and better public healthcare delivery

Moving to the cloud triggered operational changes and drove the need for new approaches to building, operating and managing applications. eHealth NSW recognised the need to upskill its employees to ensure success of its digital transformation and maximise the value of the cloud.

eHealth NSW worked with the AWS Training and Certification team to establish a multi-cloud stream within its Digital Academy (DA) in 2020.

With nine pillars, the DA is designed to accelerate the digital health workforce of the future by bringing eHealth NSW staff together to learn new skills, put learning into practice, collaborate and share ideas. The DA pillars include Agile, Analytics, Cloud, Cybersecurity, Human-Centered Design, Integration and Interoperability, Safety and Quality, Service Management, and 'Start with the Customer in Mind'. Each pillar seeks to uplift skills, processes and practices internally, and build a resilient workforce.

The multi-technology cloud stream was added to the DA in 2020 and has evolved in the past two years to include a series of pathways for staff to develop the necessary cloud skills. These pathways are targeted towards enterprise architects, cloud engineers, human factors

specialists, project managers, data analysts, coders, engineers, technicians, clinicians, business support, and experts in many other disciplines.

As of October 2022, the DA has delivered training on AWS to over 1500 technical and non-technical employees. This included a variety of foundational and in-depth courses, self-paced digital training resources, as well as experiential learning events like Lunch and Learns, AWS Immersion Days and Game Days held virtually and in person.

eHealth NSW aims to train a majority of its 2000 staff by 2023, thus equipping its employees with the right skills and confidence to build and manage applications on AWS. The DA has increased the speed at which people learn new skills and opens new cloud career opportunities for diverse learners.

"This mass reskilling pathway is not only providing new skills to ensure staff have long-term viability in the tech industry, but it is also empowering women to be more confident in tech," shared Heather Cardin, Head of Training at eHealth NSW. "It has created a community of shared learning and enablement and improved employee satisfaction."

Looking ahead, eHealth NSW will use the agility and scalability that the cloud offers to advance its capabilities in analytics, artificial intelligence and machine learning to deliver predictive and personalised healthcare services to the citizens of NSW.